

Natural Language Processing

5. Vector semantics and sparse embeddings

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Notice

Credits

The present slides are largely an adaptation of existing material, including slides by [Jurafsky & Martin](#), [Manning](#), [Potts & MacCartney](#), [Black & Mortensen](#), and others, to whom I am very grateful.

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Vector Semantics and Embeddings

What Is a *Pipe*?



"The famous pipe. How people reproached me for it! And yet, could you stuff my pipe? No, it's just a representation, is it not? So if I had written on my picture 'This is a pipe', I'd have been lying!"
(René Magritte)

Lexical Semantics

The screenshot shows the Merriam-Webster Dictionary website. The search bar at the top contains the word "pipe". The left sidebar shows the "Dictionary" tab selected, with a list of options: Definition, noun (highlighted), verb, Synonyms, Example Sentences, Word History, Phrases Containing, Related Articles, Entries Near, and Show More. The main content area displays the entry for "pipe", which is a noun. It includes the plural form "pipes", a link to synonyms, and a list of senses. Sense 1a is a tubular wind instrument, with sub-senses for a small fipple flute (fipple) and one of the tubes of a pipe organ. Sense 1b is a means of transmission (as of television signals or computer data). Sense 2a is a long tube or hollow body for conducting a liquid, gas, or finely divided solid or for structural purposes. Sense 2b is a means of transmission (as of television signals or computer data).

pipe 1 of 2 **noun**

noun

plural pipes

Synonyms of pipe

1 a : a tubular wind instrument

specifically : a small fipple flute held in and played by the left hand

b : one of the tubes of a pipe organ:

(1) : **FLUE PIPE**

(2) : **REED PIPE**

c : **BAGPIPE** → usually used in plural

d (1) pipes : **VOCAL CORDS, VOICE**

... the actress famously showed off her *pipes* with the emotional ballad "There Are Worse Things I Could Do."

— Sophie Dodd

a singer with a great *set of pipes*

(2) : **PIPING sense 1**

2 a : a long tube or hollow body for conducting a liquid, gas, or finely divided solid or for structural purposes

b : a means of transmission (as of television signals or computer data)

a broadband fiber-optic *pipe*

Merriam Webster Dictionary, retrieved online (2024)

Lemmas and Senses

- The form *pipe* is the **lemma** or **citation form** for pipe and pipes (and pype, pypes, pyppe, ...)
 - Dictionaries don't have separate definitions for inflected or archaic forms
- A **word sense** or **concept** is the meaning component of a word
 - Lemmas can be **polysemous**
 - **Word sense disambiguation**: determining the correct sense in context

Synonymy

- A relation of (near) identity between two senses of two different words
 - *couch/sofa, water/H₂O, car/automobile*
 - words are synonyms if mutually substitutable for one another in any sentence without changing the truth condition
 - same **propositional meaning**
- However, **principle of contrast**: a difference in linguistic form is always associated with some difference in meaning
 - probably no examples of perfect synonymy
 - politeness, slang, register, genre, etc.

Pipe Synonyms

pipe 1 of 2 **noun**

'pīp «»

Definition of *pipe* >

Get Custom Synonyms ↗

Enter your own sentence containing **pipe**, and get words to replace it.



1 as in *conduit*

a long hollow cylinder for carrying a substance (as a liquid or gas)
the plumber came and fixed the water *pipe* that was leaking

Synonyms & Similar Words

Relevance

conduit

tube

channel

funnel

pipng

drain

duct

line

trough

penstock

pipeline

leader

Merriam Webster Dictionary, retrieved online (2024)

Pipe Synonyms

Get Custom Synonyms ✨

I opened a pipe and sent my message with it

→

Confidence

.....

I opened a **tube** and sent my message with it

I opened a **line** and sent my message with it

I opened a **channel** and sent my message with it

Merriam Webster Dictionary, retrieved online (2024)

Antonymy and Other Relations

- Senses that are opposites with respect with one feature of meaning
 - otherwise, they are very similar!
 - *dark/light, short/long, fast/slow, in/out, rise/fall*
- More formally: antonyms can
 - define a binary opposition or be at opposite ends of a scale (*long/short, fast/slow*)
 - be *reversives* (*rise/fall, up/down*)
- Other relations among senses:
 - **subordination, hyponymy** or **IS-A**: *car* is a subordinate of *vehicle*
 - **superordination** or **hypernymy**: *furniture* is a superordinate of *lamp*, which is a superordinate of *desk lamp*
 - **meronymy** (part-whole)

Graph Visualization of Relations Among Word Senses

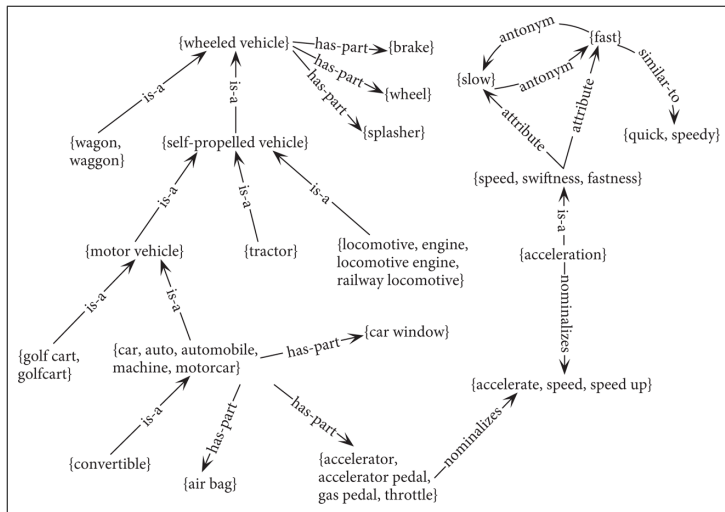


Figure G.6 WordNet viewed as a graph. Figure from Navigli (2016).

WordNet

WordNet Search - 3.1

- [WordNet home page](#) - [Glossary](#) - [Help](#)

Word to search for:

Display Options:

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations

Display options for sense: (gloss) "an example sentence"

Noun

- [S:](#) (n) **pipe**, [tobacco pipe](#) (a tube with a small bowl at one end; used for smoking tobacco)
- [S:](#) (n) **pipe**, [pipage](#), [piping](#) (a long tube made of metal or plastic that is used to carry water or oil or gas etc.)
- [S:](#) (n) **pipe**, [tube](#) (a hollow cylindrical shape)
- [S:](#) (n) **pipe** (a tubular wind instrument)
- [S:](#) (n) [organ pipe](#), **pipe**, [pipework](#) (the flues and stops on a pipe organ)

Verb

- [S:](#) (v) [shriek](#), [shrill](#), [pipe up](#), **pipe** (utter a shrill cry)
- [S:](#) (v) **pipe** (transport by pipeline) "*pipe oil, water, and gas into the desert*"
- [S:](#) (v) **pipe** (play on a pipe) "*pipe a tune*"
- [S:](#) (v) **pipe** (trim with piping) "*pipe the skirt*"

Supersenses in WordNet

Supersenses act as coarse semantic categories. Example for nouns:

Category	Example	Category	Example	Category	Example
ACT	<i>service</i>	GROUP	<i>place</i>	PLANT	<i>tree</i>
ANIMAL	<i>dog</i>	LOCATION	<i>area</i>	POSSESSION	<i>price</i>
ARTIFACT	<i>car</i>	MOTIVE	<i>reason</i>	PROCESS	<i>process</i>
ATTRIBUTE	<i>quality</i>	NATURAL EVENT	<i>experience</i>	QUANTITY	<i>amount</i>
BODY	<i>hair</i>	NATURAL OBJECT	<i>flower</i>	RELATION	<i>portion</i>
COGNITION	<i>way</i>	OTHER	<i>stuff</i>	SHAPE	<i>square</i>
COMMUNICATION	<i>review</i>	PERSON	<i>people</i>	STATE	<i>pain</i>
FEELING	<i>discomfort</i>	PHENOMENON	<i>result</i>	SUBSTANCE	<i>oil</i>
FOOD	<i>food</i>			TIME	<i>day</i>

Figure G.2 Supersenses: 26 lexicographic categories for nouns in WordNet.

Noun Relations in WordNet

Relation	Also Called	Definition	Example
Hypernym	Superordinate	From concepts to superordinates	<i>breakfast</i> ¹ → <i>meal</i> ¹
Hyponym	Subordinate	From concepts to subtypes	<i>meal</i> ¹ → <i>lunch</i> ¹
Instance Hypernym	Instance	From instances to their concepts	<i>Austen</i> ¹ → <i>author</i> ¹
Instance Hyponym	Has-Instance	From concepts to their instances	<i>composer</i> ¹ → <i>Bach</i> ¹
Part Meronym	Has-Part	From wholes to parts	<i>table</i> ² → <i>leg</i> ³
Part Holonym	Part-Of	From parts to wholes	<i>course</i> ⁷ → <i>meal</i> ¹
Antonym		Semantic opposition between lemmas	<i>leader</i> ¹ ⇔ <i>follower</i> ¹
Derivation		Lemmas w/same morphological root	<i>destruction</i> ¹ ⇔ <i>destruction</i> ¹

Figure G.3 Some of the noun relations in WordNet.

Verb Relations in WordNet

Relation	Definition	Example
Hypernym	From events to superordinate events	<i>fly</i> ⁹ → <i>travel</i> ⁵
Troponym	From events to subordinate event	<i>walk</i> ¹ → <i>stroll</i> ¹
Entails	From verbs (events) to the verbs (events) they entail	<i>snore</i> ¹ → <i>sleep</i> ¹
Antonym	Semantic opposition between lemmas	<i>increase</i> ¹ ⇔ <i>decrease</i> ¹

Figure G.4 Some verb relations in WordNet.

Word Sense Disambiguation

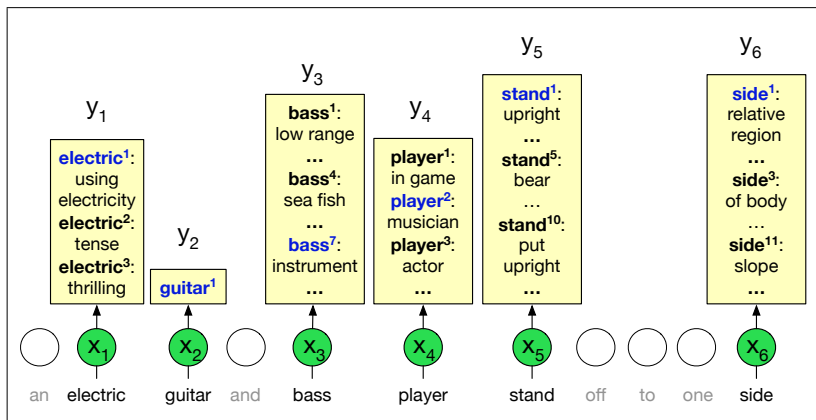


Figure G.8 The all-words WSD task, mapping from input words (x) to WordNet senses (y). Only nouns, verbs, adjectives, and adverbs are mapped, and note that some words (like *guitar* in the example) only have one sense in WordNet. Figure inspired by [Chaplot and Salakhutdinov \(2018\)](#).

Word Similarity

- Words have few synonyms, but lots of *similar* words
 - *cat/dog*
 - not synonyms, but certainly have **similar meaning**
- shifting from relation between senses to relation between words
simplifies things
- similarity is *useful* in larger semantic tasks
 - sentence similarity
 - question-answering
 - paraphrasing
 - summarization
- Human judgment to build datasets like **SimLex-999**

9.8	<i>vanish/disappear</i>	5.95	<i>belief/impression</i>	0.98	<i>modest/flexible</i>
7.3	<i>behave/obey</i>	3.65	<i>muscle/bone</i>	0.3	<i>hole/agreement</i>

Word Relatedness/Association

- Any relation, perhaps via a semantic frame or field
 - *car/bicycle, coffee/tea, medic/surgeon*: **similar**
 - *car/fuel, coffee/cup, scalpel/surgeon*: **related**, not similar
- **Semantic field**: words that cover a particular domain and bear structured relations with each other
 - hospitals:
 - *surgeon, scalpel, nurse, anaesthetic, hospital*
 - restaurants:
 - *waiter, menu, plate, food, menu, chef*
 - houses:
 - *door, roof, kitchen, family, bed*
- Related to **topic models** and **semantic frames**

Topic Models

“Arts”	“Budgets”	“Children”	“Education”
NEW	MILLION	CHILDREN	SCHOOL
FILM	TAX	WOMEN	STUDENTS
SHOW	PROGRAM	PEOPLE	SCHOOLS
MUSIC	BUDGET	CHILD	EDUCATION
MOVIE	BILLION	YEARS	TEACHERS
PLAY	FEDERAL	FAMILIES	HIGH
MUSICAL	YEAR	WORK	PUBLIC
BEST	SPENDING	PARENTS	TEACHER
ACTOR	NEW	SAYS	BENNETT
FIRST	STATE	FAMILY	MANIGAT
YORK	PLAN	WELFARE	NAMPHY
OPERA	MONEY	MEN	STATE
THEATER	PROGRAMS	PERCENT	PRESIDENT
ACTRESS	GOVERNMENT	CARE	ELEMENTARY
LOVE	CONGRESS	LIFE	HAITI

The William Randolph Hearst Foundation will give \$1.25 million to Lincoln Center, Metropolitan Opera Co., New York Philharmonic and Juilliard School. “Our board felt that we had a real opportunity to make a mark on the future of the performing arts with these grants an act every bit as important as our traditional areas of support in health, medical research, education and the social services,” Hearst Foundation President Randolph A. Hearst said Monday in announcing the grants. Lincoln Center’s share will be \$200,000 for its new building, which will house young artists and provide new public facilities. The Metropolitan Opera Co. and New York Philharmonic will receive \$400,000 each. The Juilliard School, where music and the performing arts are taught, will get \$250,000. The Hearst Foundation, a leading supporter of the Lincoln Center Consolidated Corporate Fund, will make its usual annual \$100,000 donation, too.

Latent Dirichlet Allocation, Blei et al, JMLR 2003

Semantic Frames

- A set of words that denote perspectives or participants in a particular type of event
 - For example: *commercial transaction*, where one entity trades money to another entity in return for some good or service, after which the good changes hands or the service is performed
 - lexical encoding: verbs like *buy*, *sell*, *pay*, or nouns like *buyer*
 - frames have **semantic roles** (like *buyer*, *seller*, *goods*, *money*)
 - words can take on these roles
- Important for various tasks: **paraphrasing**, **question answering**, **machine translation**

Semantic Role Labeling

InVeRo - SRL

API DOC DOWNLOAD ABOUT

TAG

[List of Semantic Roles](#)
[List of Frames](#)

Selected source:

VerbAtlas

Sally

cut

the

banana

with

a

knife

AGENT

CUT

PATIENT

INSTRUMENT



SAPIENZA
UNIVERSITÀ DI ROMA



elexis
european lexicographic
infrastructure



Babelscape



ELG





erc

VerbAtlas is funded by the MOUSSE ERC Grant no.726487 and the ELEXIS project no.731015 under the European Union's Horizon 2020 research and innovation programme.

InVeRo-SRL is part of the Universal Semantic Annotator (USaA) project funded by the European Language Grid (ELG), Project Number 826627.

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Screenshot from [Verbatlas.org](https://www.verbatlas.org/)

Connotation

- Affective meaning, sentiment
- Remember the dimensions of affective meaning?

	Valence	Arousal	Dominance
courageous	8.05	5.5	7.38
music	7.67	5.57	6.5
heartbreak	2.45	5.65	3.58
cub	6.71	3.95	4.24
life	6.68	5.59	5.89

- A revolutionary idea: **word meaning could be represented as a point in space**
- The first expression of **vector semantics models**

The measurement of meaning, C Osgood et al, 1957

Different Aspects of Meaning

- Different aspects of word meaning: word senses, similarity and relatedness, semantic fields and frames, connotation, etc.
- Perfect model that completely deals with each of them seems elusive
- Example: what is the meaning of “game”?

Wittgenstein: Philosophical Investigations (1945)

66. *Consider for example [...] "games". I mean board-games, card-games, ball-games, Olympic games, and so on. What is common to them all? [...] if you look at them you will not see something that is common to all, but similarities, relationships, and a whole series of them at that [...]. Look for example at board-games [...]. Now pass to card-games [...] to ball-games, much that is common is retained, but much is lost. Are they all 'amusing'? [...] Or is there always winning and losing, or competition between players? [...] In ball games there is winning and losing; but when a child throws his ball at the wall and catches it again, this feature has disappeared. [...] we see a complicated network of similarities overlapping and criss-crossing: sometimes overall similarities, sometimes similarities of detail.*

67. *I can think of no better expression to characterize these similarities than "family resemblances"; for the various resemblances between members of a family: build, features, colour of eyes, gait, temperament, etc. etc. overlap and criss-cross in the same way. [...]*

Vector Semantics

- Alternative view: words defined by their usage

“The meaning of a word is its use in the language”

(Wittgenstein, 1953)

- Linguistic distributionalists
 - Define a word by its environment or *distribution* in language use
 - **Distribution**: the set of contexts in which it occurs, the neighbouring words or grammatical environments
 - very similar distributions → likely the same meaning

“If A and B have almost identical environments we say that they are synonyms.”

(Harris, 1954)

“A word is characterized by the company it keeps.” (Firth, 1950)

Distributional structure, Z Harris (1954)

A synopsis of linguistic theory 1930-1955, JR Firth (1957)

What's Ongchoi?

- Suppose you see these sentences:
 - Onhchoi is delicious sautéed with garlic
 - Ongchoi is superb over rice
 - Ongchoi leaves with salty sauces
- And you've also seen these:
 - ...spinach sautéed with garlic over rice
 - Chard stems and leaves are delicious
 - Collard greens and other salty leafy greens
- Conclusion:
 - Ongchoi is a leafy green like spinach, chard, or collard greens

Which one is Ongchoi?



Vector Semantics

- Combination of two intuitions:
 - Distributionalist intuition: define a word by counting neighbouring words
 - Vector intuition: defining the meaning of a word as a point in an N-dimensional space (Osgood et al)



Figure 6.1 A two-dimensional (t-SNE) projection of embeddings for some words and phrases, showing that words with similar meanings are nearby in space. The original 60-dimensional embeddings were trained for sentiment analysis. Simplified from [Li et al. \(2015\)](#) with colors added for explanation.

Embeddings

- Each word = a vector
 - “Embedding” because it’s embedded into a space
 - Not just “word” or w_{45}
 - Similar words are **nearby in space**
- The standard way to represent meaning in NLP
- Embeddings enable computing **similarity** between two words, sentences, documents
 - Useful in many NLP tasks like sentiment analysis
 - With words, requires **same** word to be in training and test
 - With embeddings: ok if **similar** words occurred!!!
 - Question answering, conversational agents, etc
- **Can be learned automatically** from text without any complex labeling or supervision

Two Common Models of Embeddings

TF-IDF

- A common baseline model
- Sparse vectors
- Words are represented by a simple function of the counts of nearby words

Word2vec

- Dense vectors
- Representation is created by training a classifier to distinguish nearby and far-away words

Words and Vectors

Words as Vectors

- Vector or distributional models of meaning are generally based on a **co-occurrence matrix**, a way of representing how often words co-occur.
- Co-occurrence can be formalized in several ways, and so do co-occurrence matrices
- Different design choices induce different word vectors

Many Design Choices

- **Matrix design**
- **Reweighting**
- **Dimensionality reduction**
- **Vector comparison**

Many Design Choices

- **Matrix design**

- Word $\times$ word
- Word $\times$ document
- Word $\times$ discourse context
- Word $\times$ search query
- Word $\times$ dependency relation
- Word $\times$ person
- Adjective $\times$ modified noun
- Phonological segment $\times$ feature values
- Verb $\times$ subject $\times$ object
- ...

- **Reweighting**

- **Dimensionality reduction**

- **Vector comparison**

Many Design Choices

- **Matrix design**
- **Reweighting**
 - Probabilities
 - Length normalization
 - TF-IDF
 - PMI
 - Positive PMI
 - ...
- **Dimensionality reduction**
- **Vector comparison**

Many Design Choices

- **Matrix design**
- **Reweighting**
- **Dimensionality reduction**
 - Latent Semantic Analysis
 - Non-negative Matrix Factorization
 - Probabilistic LSA
 - Latent Dirichlet Allocation
 - Principal Component Analysis
 - t-SNE
 - ...
- **Vector comparison**

Many Design Choices

- **Matrix design**
- **Reweighting**
- **Dimensionality reduction**
- **Vector comparison**
 - Euclidean
 - Cosine
 - Dice
 - Jaccard
 - KL divergence
 - ...

Co-occurrence Matrix

- A **co-occurrence matrix** represents how often words co-occur

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

- Example: $|V| \times |D|$ **term-document matrix**, part of vector-space model of **information retrieval**

Co-occurrence Matrix

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Figure 6.3 The term-document matrix for four words in four Shakespeare plays. The red boxes show that each document is represented as a column vector of length four.

- Example: $|V| \times |D|$ **term-document matrix**, part of vector-space model of **information retrieval**
 - A document is a count vector, identifying a point in a V -dimensional space

Co-occurrence Matrix

- A **co-occurrence matrix** represents how often words co-occur
- Example: $|V| \times |D|$ **term-document matrix**, part of vector-space model of **information retrieval**

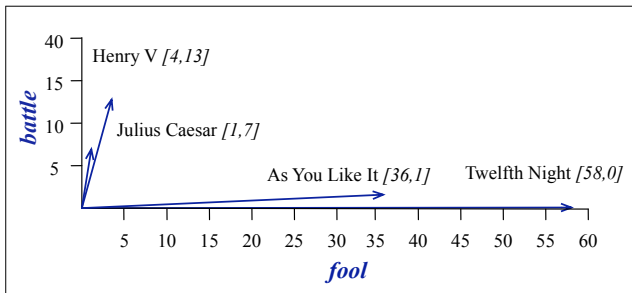


Figure 6.4 A spatial visualization of the document vectors for the four Shakespeare play documents, showing just two of the dimensions, corresponding to the words *battle* and *fool*. The comedies have high values for the *fool* dimension and low values for the *battle* dimension.

Words as Vectors

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
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Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

- Vector representation of documents
 - **As You Like It** is document [1, 114, 36, 20]
- **Words can be vectors too**
 - **battle** is “the kind of word that occurs in Julius Caesar and Henry V”
 - **fool** is “the kind of word that occurs in comedies, especially Twelfth Night”

Words as Vectors

- More commonly, **word-word matrix**
- $|V| \times |V|$ matrix,
 - rows are labeled by (target) words
 - columns are labeled by (context) words, rather than docs

is traditionally followed by	cherry	pie, a traditional dessert
often mixed, such as	strawberry	rhubarb pie. Apple pie
computer peripherals and personal	digital	assistants. These devices usually
a computer. This includes	information	available on the internet

- In this example, the context is 4 words on each side



For fun: **play with NLTK's concordance**

Words as Vectors

- Which words are similar?

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	
strawberry	0	...	0	0	1	60	19	
digital	0	...	1670	1683	85	5	4	
information	0	...	3325	3982	378	5	13	

Figure 6.5 Co-occurrence vectors for four words in the Wikipedia corpus, showing six of the dimensions (hand-picked for pedagogical purposes). The vector for *digital* is outlined in red. Note that a real vector would have vastly more dimensions and thus be much sparser.

- Notice: $|V|$ is size of vocabulary, say 10K-50K words
- Keeping words after the most frequent 50K or so may be not helpful
- Need efficient handling of **sparse** representations

Visualization

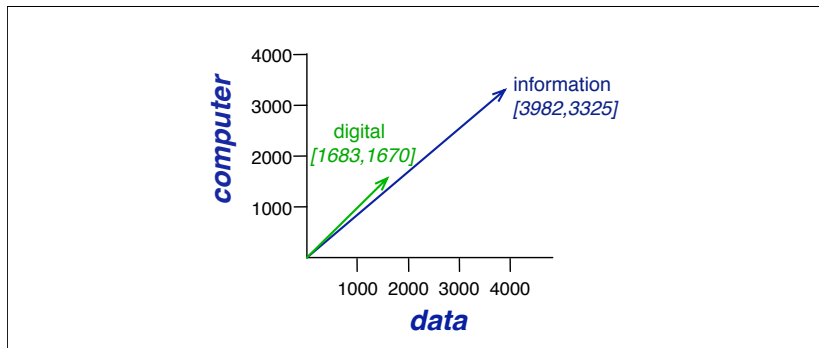


Figure 6.6 A spatial visualization of word vectors for *digital* and *information*, showing just two of the dimensions, corresponding to the words *data* and *computer*.

Windows and Scaling: What Is a Co-occurrence?

	is	then	followed	by	cherry	pie	,	a	dessert	whose
	4	3	2	1	0	1	2	3	4	5
Window: 3	is	<u>then</u>	<u>followed</u>	<u>by</u>	<u>cherry</u>	<u>pie</u>	,	<u>a</u>	dessert	whose
Scaling: flat	0	1	1	1	1	1	1	1	0	
Scaling: $\frac{1}{n}$	0	$\frac{1}{3}$	$\frac{1}{2}$	1	1	1	$\frac{1}{2}$	$\frac{1}{3}$	0	

- Larger, flatter windows capture more semantic information.
- Small, more scaled windows capture more syntactic (collocational) information.

Vector Comparison

- **dot-product**($\mathbf{w}, \mathbf{v}$) = $\mathbf{w} \cdot \mathbf{v} = \sum_{i=1}^n w_i v_i$
- Good similarity metric
 - High value if $\mathbf{w}$ and $\mathbf{v}$ have large values in the same dimensions
 - 0 if $\mathbf{w}$ and $\mathbf{v}$ orthogonal
- **length**($\mathbf{v}$) = $|\mathbf{v}| = \sqrt{\sum_{i=1}^n v_i^2}$
- **cosine**($\mathbf{w}, \mathbf{v}$) = $\frac{\mathbf{w} \cdot \mathbf{v}}{|\mathbf{w}| |\mathbf{v}|}$ (normalized dot product)

Visualization

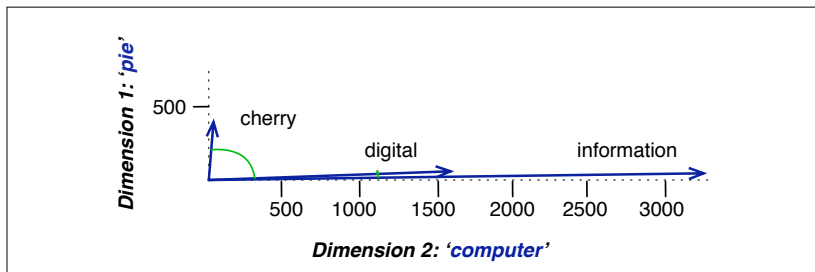


Figure 6.7 A (rough) graphical demonstration of cosine similarity, showing vectors for three words (*cherry*, *digital*, and *information*) in the two dimensional space defined by counts of the words *computer* and *pie* nearby. Note that the angle between *digital* and *information* is smaller than the angle between *cherry* and *information*. When two vectors are more similar, the cosine is larger but the angle is smaller; the cosine has its maximum (1) when the angle between two vectors is smallest (0°); the cosine of all other angles is less than 1.

A Frequency Paradox

- Frequency is useful (as seen)
- However, overly frequent words (like stopwords) are not very informative about the context
- Raw frequency is a bad representation: skewed and not very discriminative

Reweighting

Goals of Reweighting

- Amplify the important, the trustworthy, the unusual; deemphasize the mundane and the quirky
- The intuition behind moving away from raw counts is that frequency is a poor proxy for the above values
- We hope to do no feature selection based on counts, stopword dictionaries, etc.
- Rather, we want our methods to reveal what's important without these ad hoc interventions

$$\text{tf-idf}_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

- **tf: term frequency:** frequency count (usually log-transformed)

$$\text{tf}_{t,d} = \begin{cases} 1 + \log_{10} \text{Count}(t, d) & \text{if } \text{Count}(t, d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Words that occur many times in a document have a high tf
- **idf: inverse document frequency**

$$\text{idf}_t = \log_{10} \frac{N}{\text{df}_t}$$

- Words that occur only in a few documents have a high idf
- df_t is the number of documents where t occurs
- Words like *the* or *good* have a low idf

Discriminative Words in Shakespeare's Plays

Word	df	idf
Romeo	1	1.57
salad	2	1.27
Falstaff	4	0.967
forest	12	0.489
battle	21	0.246
wit	34	0.037
fool	36	0.012
good	37	0
sweet	37	0

TF-IDF Weighted Term-Document Matrix

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.246	0	0.454	0.520
good	0	0	0	0
fool	0.030	0.033	0.0012	0.0019
wit	0.085	0.081	0.048	0.054

Figure 6.9 A portion of the tf-idf weighted term-document matrix for four words in Shakespeare plays, showing a selection of 4 plays, using counts from Fig. 6.2. For example the 0.085 value for *wit* in *As You Like It* is the product of $tf = 1 + \log_{10}(20) = 2.301$ and $idf = .037$. Note that the idf weighting has eliminated the importance of the ubiquitous word *good* and vastly reduced the impact of the almost-ubiquitous word *fool*.

Using Tf-Idf to Compare Documents

- **Word similarity:** tf-idf cosine
 - For example, find 10 most similar words to **w** by computing cosine between **w** and all other $|V| - 1$ words, and looking at the top 10
- **Document similarity:**
 - Centroid (multidimensional mean) of vectors of all k words in the document

$$d = \frac{\sum_i w_i}{k}$$

- Many applications of document similarity, for example:
 - information retrieval
 - plagiarism detection
 - news recommender system
 - digital humanities

Positive Pointwise Mutual Information

- An alternative to tf-idf

To what extent are w_1 and w_2 associated?

$\Leftrightarrow$

How much **more** do w_1 and w_2 co-occur than by chance?

- Pointwise Mutual Information: $PMI(w, c) = \log_2 \frac{P(w, c)}{P(w)P(c)}$
 - $P(w, c)$ is the probability of co-occurrence
 - $P(w)P(c)$ is the probability of chance co-occurrence
- PMI's range is $(-\infty, \infty)$
- Focus on similarity: not much value in distance $\Rightarrow$ take positive part

$$PPMI(w, c) = \max(PMI(w, c), 0)$$

PPMI Weighted Term-Document Matrix

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	
strawberry	0	...	0	0	1	60	19	
digital	0	...	1670	1683	85	5	4	
information	0	...	3325	3982	378	5	13	

Figure 6.5 Co-occurrence vectors for four words in the Wikipedia corpus, showing six of the dimensions (hand-picked for pedagogical purposes). The vector for *digital* is outlined in red. Note that a real vector would have vastly more dimensions and thus be much sparser.

PPMI Weighted Term-Document Matrix

	computer	data	result	pie	sugar	count(w)
cherry	2	8	9	442	25	486
strawberry	0	0	1	60	19	80
digital	1670	1683	85	5	4	3447
information	3325	3982	378	5	13	7703
count(context)	4997	5673	473	512	61	11716

Figure 6.9 Co-occurrence counts for four words in 5 contexts in the Wikipedia corpus, together with the marginals, pretending for the purpose of this calculation that no other words/contexts matter.

- $P(\text{information}, \text{data}) = \frac{3982}{11716} = .3399$
- $P(\text{information}) = \frac{7703}{11716} = .6575$ $P(\text{data}) = \frac{5673}{11716} = .4872$
- $PPMI(\text{information}, \text{data}) = \log_2 \frac{.3399}{.6575 \times .4872} = .0944$

PPMI Weighted Term-Document Matrix

	computer	data	result	pie	sugar
cherry	0	0	0	4.38	3.30
strawberry	0	0	0	4.10	5.51
digital	0.18	0.01	0	0	0
information	0.02	0.09	0.28	0	0

Figure 6.11 The PPMI matrix showing the association between words and context words, computed from the counts in Fig. 6.10. Note that most of the 0 PPMI values are ones that had a negative PMI; for example $\text{PMI}(\text{cherry}, \text{computer}) = -6.7$, meaning that *cherry* and *computer* co-occur on Wikipedia less often than we would expect by chance, and with PPMI we replace negative values by zero.

- $P(\text{information}, \text{data}) = \frac{3982}{11716} = .3399$
- $P(\text{information}) = \frac{7703}{11716} = .6575$ $P(\text{data}) = \frac{5673}{11716} = .4872$
- $\text{PPMI}(\text{information}, \text{data}) = \log_2 \frac{.3399}{.6575 \times .4872} = .0944$

Weighting PPMI

- PMI is biased towards infrequent events
- Very rare words have very high PMI values
- Two solutions:
 - Use add- k smoothing (typically, k between .1 and 3)
 - Give rare words slightly higher probabilities

$$PPMI_{\alpha}(w, c) = \max \left(\log_2 \frac{P(w, c)}{P(w)P_{\alpha}(c)}, 0 \right)$$

$$P_{\alpha}(c) = \frac{Count(c)^{\alpha}}{\sum_c Count(c)^{\alpha}}$$

- This works because $P_{\alpha}(c) > P(c)$ for rare c
- Typically, $\alpha = .75$

Summary of Reweighting

- TF-IDF and PPMI are among the commonest weighting schemes
- General idea is: we want to weight a cell value X_{ij} relative to the value we expect given X_{i*} and X_{*j}
- Many weighting schemes end up favoring rare events that may not be trustworthy
- The magnitude of counts can be important; $[1, 10]$ and $[1000, 10000]$ might represent very different situations; length normalizing will obscure this
- PPMI amplifies counts that are tiny relative to their rows and columns. Unfortunately, with language data, these are often noise
- TF-IDF severely punishes words appearing in many documents – it behaves oddly for dense matrices like word $\times$ word matrices

Questions?